STAT 112 : Introduction to Statistics and Probability II

LECTURE 2

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Continuous Variables and Their Probability Distributions

Reflecting on random variables encountered in the real world, we are convinced that not all random variables of interest are discrete random variables. Examples of random variables that are continuous are:

- amount of rainfall measured over time
- length of life in years
- the yield of an antibiotic in a fermentation process

The probability distribution for a discrete r.v can always be given by assigning a non-negative probability to each of the possible values, the variable may assume. In every case of course, the sum of all the probabilities that we assign must be equal to 1.

Unfortunately, the probability distribution for a continuous r.v cannot be specified in the same way. It is mathematically impossible to assign non-zero probabilities to all points on a line segment/interval while satisfying the requirement that the probabilities to all the points on a line interval while satisfying the requirement that the probabilities of the distinct possible values sum to 1. As a result, we must develop a different method to describe the probability distribution for a continuous random variable.

Definition 2.1

Let X denote any random variable. the distribution function of X, denoted by $F(x)$, is such that $F(x) = P(X \leq x)$ for $-\infty < x < \infty$

Example 2.1

Suppose that X is a discrete r.v with pmf

$$p(x) = \binom{2}{x}\left(\frac{1}{2}\right)^2, x = 0, 1, 2.$$

Obtain the cdf of X, $F(x)$ and graph it

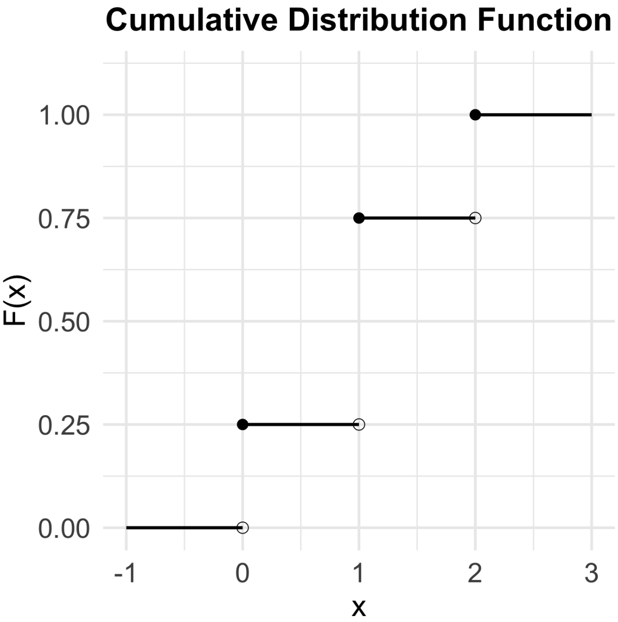
Solution

$p(0) = \frac{1}{4}, p(1) = \frac{1}{2}, p(2) = \frac{1}{4}$

$$F(x) = P(X \le x) = \begin{cases} 0, & \text{for } x < 0 \\ \frac{1}{4}, & \text{for } 0 \le x < 1 \\ \frac{3}{4}, & \text{for } 1 \le x < 2 \\ 1, & \text{for } x \ge 2 \end{cases}$$

$$\therefore \quad F(-2) = P(X \le -2) = 0$$

$$F(1.5) = P(X \le 1.5) = P(X = 0) + P(X = 1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$


Cumulative Distribution Function
F(x)
1.00
0.75
0.50
0.25
0.00
-1
0
1
2
3
x

Properties of a Distribution Function

Theorem 2.1

Given that F(x) is a distribution function, then the following properties hold:

1. $F(-\infty) = \lim_{x \to -\infty} F(x) = 0$
2. $F(\infty) = \lim_{x \to \infty} F(x) = 1$
3. $F(x)$ is a non decreasing function of x. That is if x_1 and x_2 are any values such that $x_1 < x_2$, then $F(x_1) \leq F(x_2)$.

Definition 2.2

A random variable X with distribution function $F(x)$ is continuous, for $-\infty < x < \infty$. If X is a continuous r.v, then for any real number $x, P(X = x) = 0$.

Definition 2.3

Let $F(x)$ be the distribution function for a continuous r.v X. Then $f(x)$, the pdf of X, is given by

$$f(x) = \frac{d}{dx}F(x) = F'(x)$$

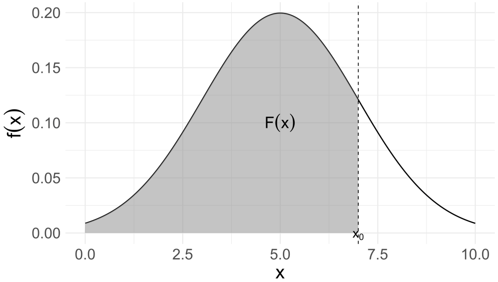
whenever the derivative exists.

It follows from Definitions 2.2 and 2.3 that F(x) can be written as

$$F(x) = \int_{-\infty}^{x} f(t)dt,$$

where $f(\cdot)$ is the pdf of X and t is used as a variable of integration.

Graphically, we have



Theorem 2.2(Properties of a Density function)

Given that $f(x)$ is a density function for a continuous r.v X, then

(i). $f(x) \geq 0$ for all x, $-\infty < x < \infty$

(ii). $\int_{-\infty}^{\infty} f(x)dx = 1$

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Example 2.2

Suppose that

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \leq x \leq 1 \\ 1 & \text{for } x > 1 \end{cases}$$

Find the pdf for X and graph it.

Solution

Since the pdf $f(x)$ is the derivative of the cdf $F(x)$, when the derivative exists, we have

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{d}{dx}(0), & \text{for } x < 0 \\ \frac{d}{dx}(x), & \text{for } 0 \leq x \leq 1 \\ \frac{d}{dx}(1), & \text{for } x > 1 \end{cases}$$

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} 1, & \text{for } 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

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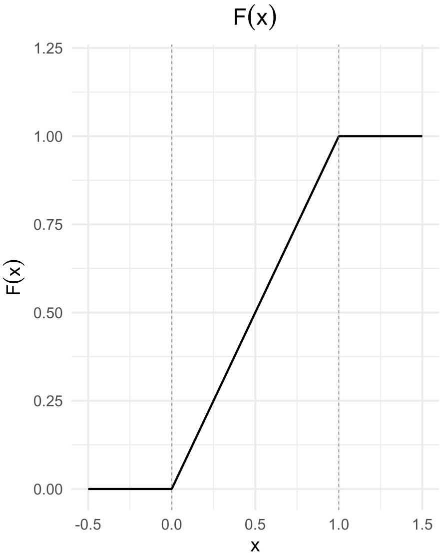
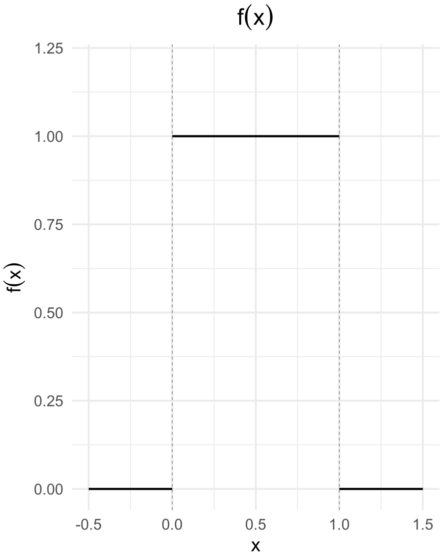

F(x)
1.25
1.00
0.75
0.50
0.25
0.00
F(x)
-0.5
0.0
0.5
1.0
1.5
x


f(x)
1.25
1.00
0.75
0.50
0.25
0.00
f(x)
-0.5
0.0
0.5
1.0
1.5
x

Example 2.3

Let X be a continuous r.v with pdf given by

$$f(x) = \begin{cases} 3x^2, & \text{for } 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

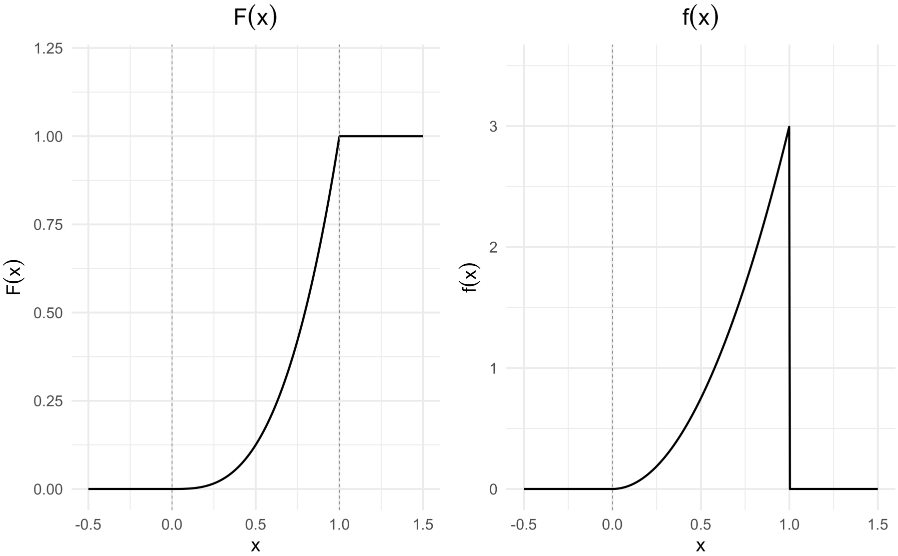
Find F(x). Graph both f(x) and F(x)

By definition,

$F(x) = \int_{\infty}^{x} f(t)dt$

$$= \begin{cases} \int_{-\infty}^{x} 0dt, & \text{for } x < 0 \\ \int_{-\infty}^{0} 0dt + \int_{0}^{x} 3t^2 dt = 0 + \left. t^3 \right|_0^x = x^3, & \text{for } 0 \leq x \leq 1 \\ \int_{-\infty}^{0} 0dt + \int_{0}^{1} 3t^2 dt + \int_{1}^{x} 0dt = 0 + \left. t^3 \right|_0^1 + 0 = 1, & \text{for } x > 1 \end{cases}$$

$$= \begin{cases} 0, & \text{for } x < 0 \\ x^3, & \text{for } 0 \leq x \leq 1 \\ 1, & \text{for } x > 1 \end{cases}$$


F(x)
f(x)
1.25
1.00
0.75
0.50
0.25
0.00
3
2
1
0
-0.5
0.0
0.5
1.0
1.5
x

Definition 2.4

Let X denote any r.v. If $0 < p < 1$, the pth quantile of X denoted γ_p, is the smallest value such that

$$P(X \leq \gamma_p) = F(\gamma_p) \geq p$$

If X is continuous, γ_p is the smallest value such that

$$F(\gamma_p) = P(X \leq \gamma_p) = p$$

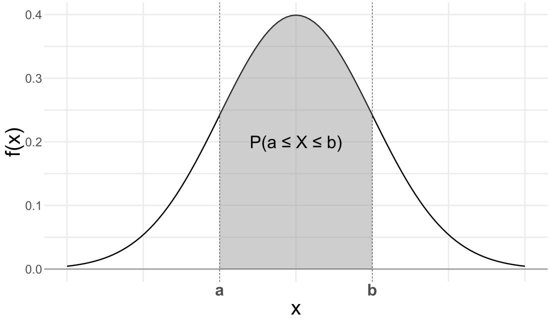
Some prefer to call γ_p, the 100pth percentile of X.

Theorem 2.3

Given that the r.v X has pdf f(x) and $a < b$, then the probability that X lies within the interval $[a, b]$ is

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

This probability is shaded in the diagram below:



If X is a continuous r.v and a and b are constants such that $a < b$, the

$P(X = a) = 0$ and $P(X = b) = 0$

Theorem 2.3 implies that

$$
\begin{aligned}
P(a < X < b) =& P(a \leq X < b) = P(a < X \leq b) \\
=& P(a \leq X \leq b) = \int_a^b f(x)dx
\end{aligned}
$$

Example 2.5

A continuous r.v X has pdf given by

$$
f(x) \begin{cases} cx^2, & \text{for } 0 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}
$$

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Find the value of c for which f(x) is a valid density function. Hence find $P(1 < X < 2)$

Solution

We obtain the value of c such that

$$F(\infty) = \int_{-\infty}^{\infty} f(x)dx = \int_{0}^{2} cx^2 dx = c \left. \frac{x^3}{3} \right|_0^2$$

$$\frac{8}{3}c = 1$$

Thus

$$c = \frac{3}{8}$$

$$\therefore f(x) = \frac{3}{8}x^2, 0 \leq x \leq 2$$

$$\therefore P(1 < X < 2) = \int_{1}^{2} f(x)dx = \frac{3}{8}\int_{1}^{2} x^2 dx = \frac{7}{8}$$

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Expected Values for Continuous Random Variables

Definition 2.5

The expected value of a continuous r.v X is given by

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

provided the integral exists.

Theorem 2.4

Let g(X) be a function of X: then the expected value of g(X) is given by

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f(x)dx$$

provided that this integral exists

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Theorem 2.5

Let c be a constant and let $g(X), g_1(X), g_2(X), \ldots, g_k(X)$ be functions of a continuous r.v X and let $c, c_1, c_2, \ldots, c_k$ be constants. Then the following results hold:

1. $E(c) = c$
2. $E[cg(X)] = cE[g(X)]$
3. $E[g_1(X)+g_2(X)+\cdots+g_k(X)] = E[g_1(X)]+E[g_2(X)]+\cdots+E[g_k(X)]$
4. $E[c_1g_1(X) + c_2g_2(X) + \cdots + c_kg_k(X)] =$
   $c_1E[g_1(X)] + c_2E[g_2(X)] + \cdots + c_kE[g_k(X)]$

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Example 2.6

Let the r.v X has density given by:

$$f(x)\begin{cases} \frac{3}{8}x^2, & \text{for } 0 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find the mean and variance of X

Solution

By definition,

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx = \int_0^2 x\left(\frac{3}{8}\right)x^2dx$$
$$= \int_0^2 \frac{3}{8}x^3 = \frac{3}{8}\left[\frac{1}{4}x^4\right]_0^2 = \frac{3}{2} = 1.5$$

To find the variance, we find

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^2 x^2 \left(\frac{3}{8}\right) x^2 dx$$

$$E(X^2) = \int_0^2 \frac{3}{8}x^4 = \frac{3}{8}\int_0^2 x^4 = \frac{3}{8}\left[\frac{1}{5}x^5\right]_0^2 = 2.4$$

$$\therefore \sigma^2 = Var(X) = E(X^2) - E^2(X)$$

$$\sigma^2 = 2.4 - 1.5^2 = 0.15$$

Example 2.7

If X is a continuous r.v with distribution function

$$F(x) = \begin{cases} 0, & \text{for } x \leq 0 \\ \frac{x}{8}, & \text{for } 0 < x < 2 \\ \frac{x^2}{16}, & \text{for } 2 \leq x < 4 \\ 1, & \text{for } x \geq 4 \end{cases}$$

Solution

Clearly the distribution function F(x) is continuous on the interval

$0 \leq x \leq \infty$, therefore, its density function, f(x) is given by

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$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{d}{dx}(0), & \text{for } x \le 0 \\ \frac{d}{dx}\left(\frac{x}{8}\right), & \text{for } 0 < x < 2 \\ \frac{d}{dx}\left(\frac{x^2}{16}\right), & \text{for } 2 \le x < 4 \\ \frac{d}{dx}(1), & \text{for } x \ge 4 \end{cases}$$

$$= \begin{cases} 0, & \text{for } x \le 0 \\ \frac{1}{8}, & \text{for } 0 < x < 2 \\ \frac{x}{8}, & \text{for } 2 \le x < 4 \\ 0, & \text{for } x \ge 4 \end{cases}$$

$$\begin{cases} 0, & \text{for } x \leq 0 \\ \frac{1}{8}, & \text{for } 0 < x < 2 \\ \frac{x}{8}, & \text{for } 2 \leq x < 4 \\ 0, & \text{elsewhere} \end{cases}$$

By definition,

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx \\ &= \int_{0}^{2} \frac{1}{8}xdx + \int_{2}^{4} \frac{x^2}{8}xdx \\ &= \frac{1}{8}\left[\frac{x^2}{2}\right]_0^2 + \frac{1}{8}\left[\frac{x^3}{3}\right]_2^4 \end{aligned}$$

$$= \frac{1}{4} + \frac{7}{3}$$

$$E(X) = \frac{3+28}{12} = \frac{31}{12}$$

Also

$$\begin{aligned}
E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x)dx \\
&= \frac{1}{8}\int_0^2 x^2 dx + \frac{x^3}{8}\int_2^4 x^3 dx \\
&= \frac{1}{8}\left[\frac{x^3}{3}\right]_0^2 + \frac{1}{8}\left[\frac{x^4}{4}\right]_2^4 \\
&= \frac{1}{8}\left(\frac{8}{3}\right) + \frac{13}{128}(64-4) \\
E(X^2) &= \frac{1}{3} + \frac{15}{2} = \frac{47}{6}
\end{aligned}$$

$$Var(X) = E(X^2) - E^2(X)$$

$$\therefore Var(X) = \frac{47}{6} - \left(\frac{31}{12}\right)^2 = \frac{167}{144} \approx 1.1597$$